

Project: Evaluating Advanced Machine Learning Models for Financial Risk Prediction using FinBench

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OBJECTIVE

To compare the performance of traditional machine learning models with the FinPT approach on the FinBench dataset for financial risk prediction. identify patterns, and explore relationships between pollution levels and various factors.

Abstract

We will do n the understanding of LLMs for financial risk prediction based on paper cite{}

DATASET

The dataset used for this project is FinBench, is a comprehensive benchmark designed for evaluating the performance of machine learning models on financial risk prediction tasks using both tabular data inputs and natural language profile text inputs. FinBench encompasses three key financial risk types as it is shown Table 1 based on (Yuwei Yin, 2023). We specifically work on Credit Default and Loan Default.

Data Exploration and Preproecsing:

The dataset contains tens of thousands of unique observations collected from real-world financial records aggregated from the Kaggle platform. The features vary by dataset but generally include demographic variables, financial indicators, and behavioral features such as account activity, transaction history, or repayment patterns. I chose ‘cd2’ and ‘ld2’ dataset which are next largest dataset after ‘ld3’ dataset so it can show the difference between Baseline models and FinPT-like model.

Table 1: Statistics of FinBench. There are three main types of financial risks, i.e., default, fraud, and churn. The default type includes two subclasses, namely credit default (cd) and loan default (ld). “cf” denotes credit fraud and “cc” means customer churn. We present the task name and description, dataset code, number of label classes, number of features, number of training/validation/test sets, and the positive (risky) sample ratio in each set.

Task	Description	Dataset	#Classes	#Features	#Train [Pos%]	#Val [Pos%]	#Test [Pos%]
Credit Default	Predict whether a user will default on the credit card or not.	cd1	2	9	2738 [7.0%]	305 [6.9%]	1305 [6.2%]
		cd2	2	23	18900 [22.3%]	2100 [22.3%]	9000 [21.8%]
Loan Default	Predict whether a user will default on the loan or not.	ld1	2	12	2118 [8.9%]	236 [8.5%]	1010 [9.0%]
		ld2	2	11	18041 [21.7%]	2005 [20.8%]	8592 [21.8%]
		ld3	2	35	142060 [21.6%]	15785 [21.3%]	67648 [22.1%]
Credit Fraud	Predict whether a user will commit fraud or not.	cf1	2	19	5352 [0.67%]	595 [1.1%]	2550 [0.90%]
		cf2	2	120	5418 [6.0%]	603 [7.3%]	2581 [6.0%]
Customer Churn	Predict whether a user will churn or not. (customer attrition)	cc1	2	9	4189 [23.5%]	466 [22.7%]	1995 [22.4%]
		cc2	2	10	6300 [20.8%]	700 [20.6%]	3000 [19.5%]
		cc3	2	21	4437 [26.1%]	493 [24.9%]	2113 [27.8%]

Table 1: Statistics of FinBench

Baseline models and FinPT-like model

We divided the dataset into 80% for training and 20% for testing, and applied baseline models suitable for both categorical and numerical data. On the *cd2* dataset, baseline models achieved metrics above 0.8. This shows that the dataset contains clear, informative features and low noise, allowing the models to easily get the underlying patterns. Additionally, the balanced distribution between classes and proper preprocessing (encoding, scaling) directed to the strong baseline performance. The FinPT-like model showed slightly lower performance compared to the other models, with metrics below 0.8 and a relatively lower AUC-ROC. This is likely because the FinPT-like architecture, originally designed for sequential or textual financial data, is less suited for structured tabular datasets like *cd2*. Additionally, the model’s higher complexity may require more data and fine-tuning to reach optimal performance which also mentioned on (Yuwei Yin, 2023).

	Accuracy	Precision	Recall	F1-score	AUC-ROC
Logistic Regression	0.821	0.806	0.821	0.799	0.772
Random Forest	0.815	0.798	0.815	0.797	0.773
Gradient Boosting	0.823	0.809	0.823	0.802	0.798
Neural Network	0.798	0.775	0.798	0.776	0.727
FinPT-like Model	0.786	0.776	0.786	0.717	0.738

Table 2: ‘cd2’ Dataset Model Performance

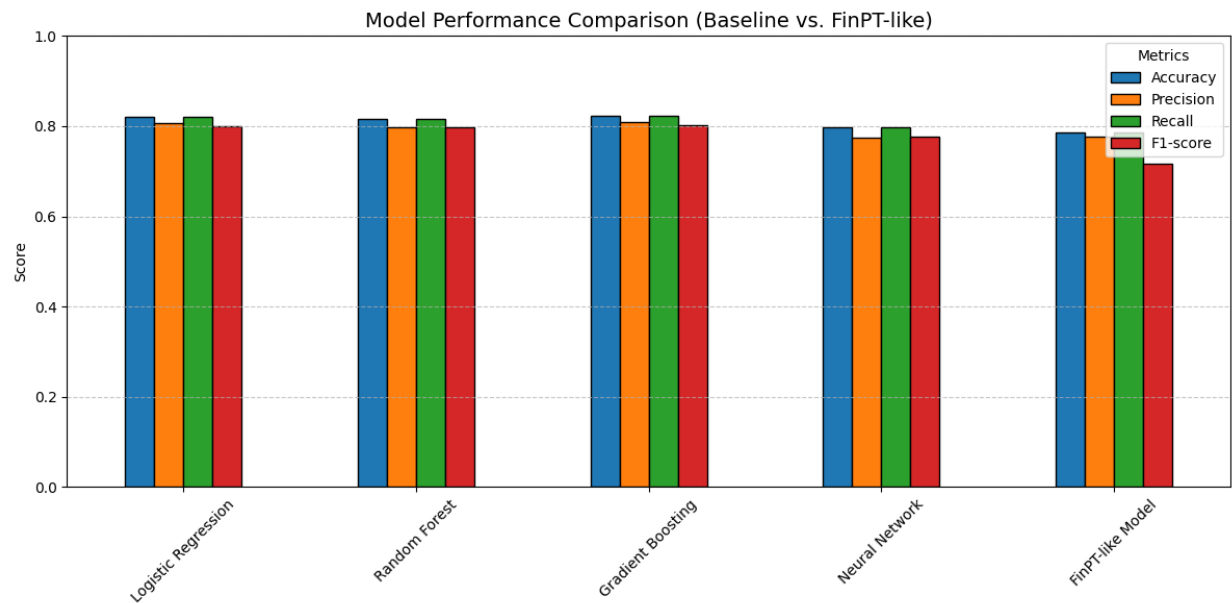


Figure 1: Line–bar chart of model performance metrics on the ‘cd2’ dataset

For the 'ld2' dataset, Random Forest, Gradient Boosting, and Neural Network performed significantly well, and there is a noticeable difference between these models and the FinPT-like model, as we can see from the line-bar chart. The FinPT-like model achieved a low F1-score. This may be because the FinPT-like model is not well-suited for tabular data and requires more data or fine-tuning to generalize effectively.

The Logistic Regression model did not perform much lower than the other baseline models. This suggests that the 'ld2' dataset may have near-linear relationships between features and target variables, allowing a simple linear model like Logistic Regression to achieve similar performance to more complex models.

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Table 3: 'ld2' Dataset Model Performance

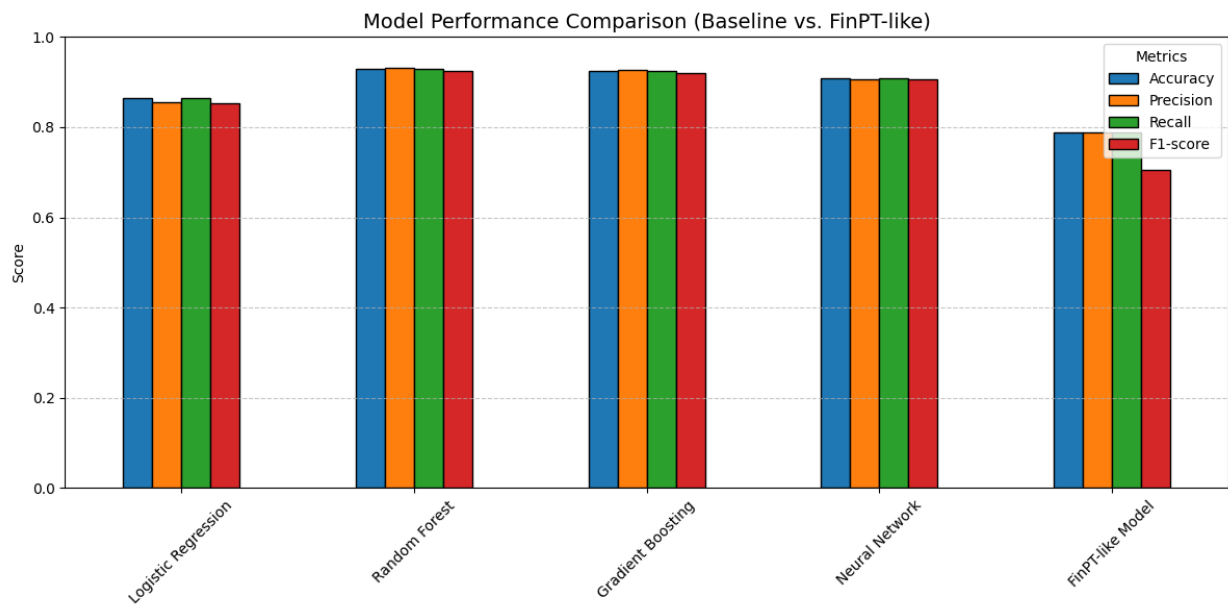


Figure 2: Line-bar chart of model performance metrics on the 'ld2' dataset.

Conclusion

For future research, the FinPT-like model could be improved by better adapting it to mixed data types, such as combining numerical and categorical features more effectively. Using financial text and market data for pretraining could also help the model learn useful patterns and generalize better. In addition, combining transformer-based models with traditional machine learning methods might create hybrid models that use the strengths of both approaches.

Bibliography

Yuwei Yin, Y. Y. (2023, July). <https://arxiv.org/abs/2308.00065>. From arXiv.